

Project-Proposal OOP



Project Domain: SOCIAL MEDIA Platform

(Connectify: A C++ Social Media Management System)

BS Software Engineering 2nd Semester (Spring)

Section: BSE-2B Course: Object Oriented Programming (OOP)

Submitted To:

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Abstract:

The **Social Media Management System** is a C++ Windows Forms (WinForms) application designed to simulate the core functionalities of a modern digital networking platform. In an era of rapid digital communication, managing complex user relationships and content flow requires efficient data structures and robust logic. This project provides a digital solution that organizes user interactions, post management, and private communication into a streamlined, automated system.

The application is built on fundamental **Object-Oriented Programming (OOP)** principles, utilizing a base Person class to derive specialized User and Admin roles. This architecture ensures **Role-Based Access Control (RBAC)**, where Admins possess elevated permissions to moderate the platform, while Users interact with the news feed and messaging systems. To ensure data persistence across sessions, the system implements **C++ File Handling (fstream)** to serialize and store user profiles, friendship graphs, and post histories. By combining the power of C++ logic with an interactive WinForms GUI, this project demonstrates a scalable approach to social software engineering.

Main Features:

1. **User Management System:** Secure Login, Signup, and Account Deletion for both Admins and Users.
2. **Admin Command Center:** High-level access to view all registered users, delete inappropriate posts, and ban users.
3. **Dynamic News Feed:** Displays posts from followed users in real-time using C++ pointers and collections.
4. **Multimedia Post Engine:** Support for creating text-based and image-based posts using Inheritance.
5. **Interactive Social Graph:** System to send, accept, or reject friend requests and manage "Followers".
6. **Engagement Module:** Real-time Liking and Commenting system on posts.
7. **Private Messaging System:** Secure one-to-one communication between connected users.
8. **Advanced Search & Discovery:** Search for users or posts using specific keywords or usernames.

9. **Persistent Data Storage:** Full system state saving and loading using fstream (File Handling).

10. **Notification Engine:** Real-time alerts for likes, comments, and new connection requests

OOP Concepts Used:

OOP Concept	Implementation in Project
Encapsulation	Using private data members for User passwords and personal info, with public "Getter/Setter" methods for secure access.
Inheritance	A base Person class derived into Admin and User classes to manage different permission levels.
Abstraction	Using abstract classes for the Post system to hide complex formatting logic from the main News Feed interface.
Polymorphism	Overriding a display() function in derived classes (TextPost, ImagePost) to show different content types in the UI.
File Handling	Using fstream to serialize user objects and post data into .txt or .dat files for persistent storage.

Expected Outcome:

The successful completion of **Connectify** will result in a fully functional, GUI-based social networking simulation that demonstrates a professional-grade command of C++ **Object-Oriented Programming**. Key outcomes include:

- **Robust System Architecture:** A scalable back-end logic where Admin and User perspectives are cleanly separated through **Inheritance** and **Role-Based Access Control**.
- **Data Integrity and Persistence:** A reliable file-handling system using fstream that ensures all user profiles, posts, and social interactions are preserved across system restarts.
- **Advanced Interaction Logic:** Implementation of complex data relationships, such as friendship graphs and real-time news feed generation using pointers and polymorphic function calls.
- **Professional Documentation and Version Control:** A complete GitHub repository featuring clean, modular code and a comprehensive README, simulating a real-world software development lifecycle.

Conclusion:

The **Social Media Management System (Connectify)** serves as a comprehensive bridge between theoretical OOP concepts and practical software engineering. By tackling the complexities of user relationships, content moderation, and data persistence, our team will gain invaluable experience in building interactive, user-centric applications. This project not only reinforces our understanding of encapsulation, polymorphism, and abstraction but also demonstrates how structured software design can solve modern communication challenges. Ultimately, Connectify stands as a testament to our team's ability to collaborate, innovate, and deliver a high-quality technical solution within a professional framework.